Rethinking VLA Safety Evaluation: When 98% of Collision Avoidance is an Illusion

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Abstract: Vision-Language-Action (VLA) models report high Collision Avoidance Rates (CAR) on safety benchmarks, yet we show these metrics are contaminated by action collapse, where near-zero motor output produces zero collisions through inaction rather than avoidance. On SafeLIBERO, an SDF-augmented Pi0 achieves 91.5% CAR while 99.6% of its safe episodes exhibit end-effector (EE) displacement below 0.1 mm. The robot is frozen, not safe. We introduce displacement-filtered CAR (dfCAR), which excludes sub-threshold episodes. The bimodal displacement distribution makes dfCAR robust across a $[2, 200]$ mm threshold range. We further show that end-effector-only Signed Distance Field (SDF) supervision fails to deliver collision avoidance. At inference time, ground-truth oracle refinement reduces CAR below unrefined baselines because the SDF gradient drives unmonitored arm links into obstacles. During training, collision-free demonstrations alone account for all measured avoidance gains ($\sim$28 percentage points). SDF supervision adds no measurable benefit across the full tested range $\lambda \in \{0.005, 0.01, 0.1\}$ ($p > 0.05$ for all; $\lambda = 0.1$: 57.00% $\pm$ 2.18, 3 seeds). Displacement filtering is necessary for meaningful VLA safety evaluation. EE-only geometric supervision is fully redundant across a 20$\times$ loss-weight range, confirming that collision avoidance arises from demonstration data, not geometric supervision.

Keywords: VLA safety, collision avoidance, action collapse, signed distance fields

1 Introduction

Collision avoidance is a prerequisite for deploying Vision-Language-Action (VLA) models in physical environments [1]. The SafeLIBERO benchmark [2] provides the first systematic protocol for measuring collision avoidance in VLA manipulation. Recent methods report high Collision Avoidance Rate (CAR) through SDF-augmented training [3], constraint-based planning [4], and collision-free demonstrations [5]. Our SDF-augmented Pi0 [6] reaches 91.5% CAR across 800 episodes. The problem appears close to solved.

It is not (Figure 1). Of the 732 collision-free episodes, 99.6% exhibit displacement below 0.1 mm; TSR is 0% across all 800 episodes. The robot is frozen, not safe. The same pattern holds across all Pi0 configurations (Section 4). We term this *fake safety*: *action collapse* (near-zero actions producing zero collisions) renders reported CAR illusory, and standard evaluation cannot distinguish inaction from deliberate avoidance [7].

We present a systematic investigation of this problem across the Pi0 and Pi0.5 [6] model families on SafeLIBERO, spanning 100+ experiments and 6000+ evaluation episodes. Three findings emerge.

Fake safety is pervasive. We introduce displacement-filtered CAR (dfCAR), which excludes episodes below a displacement threshold τ_d. dfCAR collapses Pi0+A1 from 91.5% to $\approx$0% (3/732

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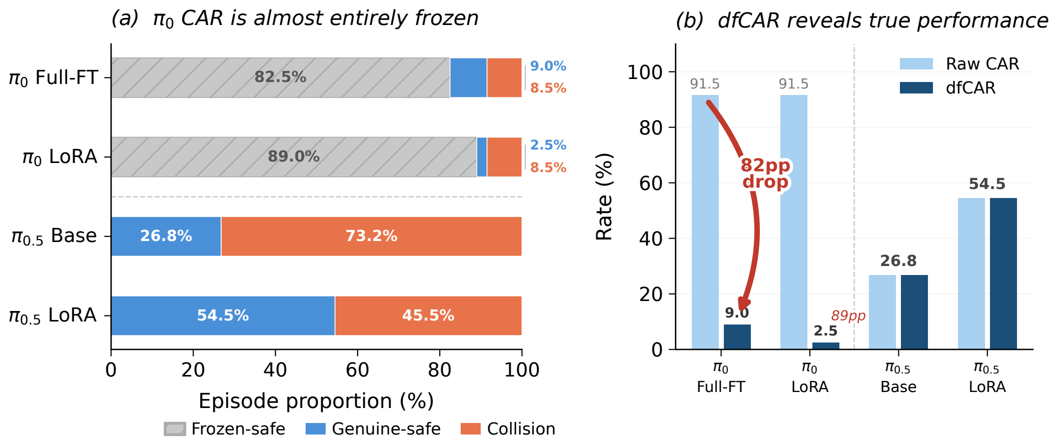

Figure 1: The fake safety problem in VLA collision avoidance evaluation. Standard CAR classifies a frozen robot as safe because it produces zero collisions (left). dfCAR filters out action-collapsed episodes where displacement falls below τ_d; Pi0+A1's 91.5% safety score drops to $\approx 0\%$ (center). Pi0.5 exhibits genuine avoidance with active motion above 288 mm in all safe episodes, and dfCAR equals raw CAR (right).

safe episodes survive). The bimodal displacement distribution makes τ_d robust across five orders of magnitude. Pi0.5 validates the metric: TSR = 71.5%, CAR = 28.5%, with every safe episode above 288 mm displacement and dfCAR equal to raw CAR (Section 4).

EE-only SDF refinement is structurally counterproductive. Three ground-truth oracle variants all achieve *lower* CAR than unrefined baselines (73.1–76.9% vs. 82–83%). The SDF gradient repels the EE but rotates arm links into obstacle volumes (Section 5).

Collision-free demonstrations are sufficient for collision avoidance. LoRA fine-tuning on Pi0.5 with collision-free demonstrations raises CAR by $\sim$26 points with zero frozen episodes. Adding SDF supervision produces no measurable benefit at low loss weights ($\lambda \leq 0.01$, $p > 0.1$) and only a marginal signal at $\lambda = 0.1$ ($p \approx 0.071$, $n = 2$). The per-task avoidance ranking is preserved across all conditions, indicating that the demonstration distribution accounts for all observed avoidance (Section 6).

Our contributions are:

- **dfCAR metric:** displacement-filtered CAR that separates genuine avoidance from action collapse, robust across $\tau_d \in [2, 200]$ mm.
- **Fake safety diagnosis:** 100+ experiments across Pi0/Pi0.5 showing >99% of Pi0 safe episodes are action-collapse artifacts.
- **EE-only SDF is counterproductive:** ground-truth oracle refinement *reduces* CAR below unrefined baselines by redistributing collisions to unmonitored arm links.
- **Demonstration sufficiency:** collision-free demonstrations account for all measured CAR gains ($\sim$26 pp); SDF supervision adds no significant benefit at $\lambda \leq 0.01$ and only a marginal signal at $\lambda = 0.1$ ($n = 2$).

2 Related Work

Vision-Language-Action Models. VLA models combine pretrained vision-language backbones with action prediction for generalist manipulation [8, 6, 9, 10]. Recent work enhances spatial reasoning through auxiliary 3D supervision [11, 12, 13], but safety considerations remain largely absent: most VLAs train on collision-free demonstrations and evaluate task success alone, with no mechanism to distinguish genuine avoidance from incidental collision-free trajectories.

Table 1: Training configurations. Trainable parameters exclude the frozen backbone.

Configuration	Description	Trainable Params
Pi0 Full-FT	Full fine-tune, flow-matching loss	$\sim$2.5B
Pi0 LoRA	LoRA adapters only	$\sim$33.5M
Pi0+A1 (SDF aux)†	LoRA + SDF decoder + EE extractor	$\sim$658K
Pi0+A3 (learned SDF)	LoRA + SDF decoder w/ gradient refinement	$\sim$34.2M
Pi0.5 (pre-trained)	No fine-tuning	0
Pi0.5 LoRA	LoRA fine-tune on Pi0.5	$\sim$33.5M
Pi0.5+A1 (SDF aux)	LoRA + SDF aux on Pi0.5, $\lambda \in \{0.005, 0.01, 0.1\}$	$\sim$34.2M

†LoRA weights not loaded due to parameter-name mismatch; only auxiliary heads trained.

Safety in Robot Learning. Safety-aware approaches span training-time constrained RL [3] and test-time filters: control barrier functions [2], collision risk gating [5], inverse kinematics QP [14], and diffusion-process collision costs [15, 4]. SafeDiffuser [16] integrates control barrier functions directly into diffusion denoising to enforce safety invariance during planning; like other test-time shields, it does not learn intrinsic collision awareness within the policy. A recent survey [1] categorizes these into runtime shields, constrained optimization, and representation-based methods. Common to all is that safety is imposed externally rather than learned within the policy's representations. This work tests whether auxiliary SDF supervision can produce intrinsic collision awareness.

Signed Distance Functions in Robotics. SDFs encode minimum surface distance, where the sign indicates free space or penetration. Neural SDFs [17] enable continuous shape representation; in robotics, SDFs serve perception [18], planning [19, 20], and differentiable collision detection [21]. CDF [22] extends SDFs to robot configuration space, learning neural distance fields that provide efficient joint-angle queries with analytical gradients for manipulation planning; CSSDF-Net [23] further develops this direction for safe motion planning. RDF [24] represents robot geometry itself as composable per-link SDFs along the kinematic chain, enabling whole-body collision queries—directly relevant to the full-body awareness gap our work identifies. Neural SDF-based model predictive control [25] demonstrates effectiveness in classical pipelines, but integration into end-to-end learned policies remains unexplored.

Evaluation Metrics for Safe Manipulation. SafeLIBERO [2] extends LIBERO [26] with obstacle-augmented environments and a collision avoidance rate (CAR) metric. However, CAR treats all collision-free episodes identically: a frozen robot scores near-perfect CAR without any avoidance behavior. This failure mode echoes concerns about metric reliability [7, 27] and binary success metrics under limited samples [28]. Action collapse in diffusion policies [29] systematically inflates CAR but has not been previously examined as a safety evaluation confounder. We propose dfCAR (§3.2) to detect this contamination.

3 Experimental Setup and Metrics

3.1 Benchmark and Models

We evaluate on SafeLIBERO [2], the *object_pudding* suite (Level II): four tabletop pick-and-place tasks with collision hazards and per-timestep collision detection. Two VLA families are tested: **Pi0** ($\sim$2.5B) [6] with flow matching [30], and **Pi0.5** with broader pre-training. Table 1 lists all configurations. Each uses 50 episodes per task (200 total); Pi0.5 multi-seed runs use 3 seeds (600 episodes). All metrics are computed using a fixed evaluation seed (seed = 7); the reported $\pm$ values reflect variance across training seeds, not evaluation stochasticity. We report Collision Avoidance Rate (CAR), the fraction of collision-free episodes, and Task Success Rate (TSR), the fraction of episodes that complete the task [2, 3].

Table 2: Fake safety decomposition on SafeLIBERO. dfCAR excludes episodes with displacement $< \tau_d = 0.1$ mm.

Method	Ep.	CAR (%)	TSR (%)	Fake Safe (%)	dfCAR (%)
Pi0 Full-FT	200	83.0	0.5	99.4	≈ 0
Pi0 LoRA	200	82.0	0.0	$\sim$99	≈ 0
Pi0+A1 (SDF)	800	91.5	0.0	99.6	$\approx 0^{*}$
Pi0.5	200	28.5	71.5	0.0	28.5

* 3/732 safe episodes above threshold; denominator too small for reliable rate.

3.2 Displacement-Filtered CAR

Standard CAR conflates genuine avoidance with action collapse. We define an episode as *action-collapsed* when $d_{\max}(e) < \tau_d$ ($\tau_d = 0.1$ mm) and exclude such episodes:

$$\text{dfCAR} = \frac{|\{e : \text{safe}(e) \ \wedge \ d_{\max}(e) \geq \tau_d\}|}{|\{e : d_{\max}(e) \geq \tau_d\}|} \tag{1}$$

The displacement distribution is sharply bimodal, with collapsed episodes below 1 μm and active episodes above 200 mm; any $\tau_d \in [2, 200]$ mm yields identical results (Appendix B).[1]

3.3 SDF Auxiliary Training and Oracle Refinement

We augment LoRA-adapted [31] VLAs with an SDF decoder and an EE position extractor (Appendix J, Table 12). The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{action}} + \lambda \left(\mathcal{L}_{\text{SDF}} + \mathcal{L}_{\text{EE}}\right) \tag{2}$$

where $\lambda \in \{0.005, 0.01, 0.1\}$. In Pi0+A1, a parameter-name mismatch prevented LoRA loading and caused action collapse (TSR = 0%; Table 1). To test the upper bound of EE-only refinement (Section 5), we also replace the learned SDF with ground-truth distance fields from the simulation geometry in three oracle variants (Appendix I).

4 The Fake Safety Problem

Standard CAR assumes that collision-free episodes reflect deliberate avoidance. This assumption is systematically violated: in every Pi0 configuration tested, over 99% of collision-free episodes are action-collapse artifacts (Table 2).[2] Applying dfCAR collapses all three to $\approx 0\%$. Pi0+A1 achieves the *highest* raw CAR (91.5%) because more complete action collapse (from a parameter-name mismatch preventing LoRA loading; Table 1) yields higher fake safety. Pi0.5 validates the metric from the opposite direction: every safe episode exceeds 200 mm displacement, and dfCAR equals raw CAR (28.5%). The displacement distribution is sharply bimodal (Appendix A), with a five-order-of-magnitude gap that makes τ_d robust.

Figure 2 reveals spatial structure in the TSR-CAR plane. Pi0-family models cluster in the upper-left (high CAR, near-zero TSR), Pi0.5 in the lower-right (high TSR, low CAR). The upper-right quadrant (high TSR *and* CAR) represents genuinely safe behavior; the upper-left is accessible through action collapse alone. Under dfCAR, Pi0 configurations drop to the origin: the apparent safety-capability tradeoff was an artifact of action-collapse contamination.

[1] See Section 4 for details on the displacement measurement protocol.

[2] Pi0-family values report obstacle displacement from the original protocol; Pi0.5 reports EE displacement from the rotation-corrected protocol (§3.2). Sub-millimeter displacement indicates action collapse in both cases.

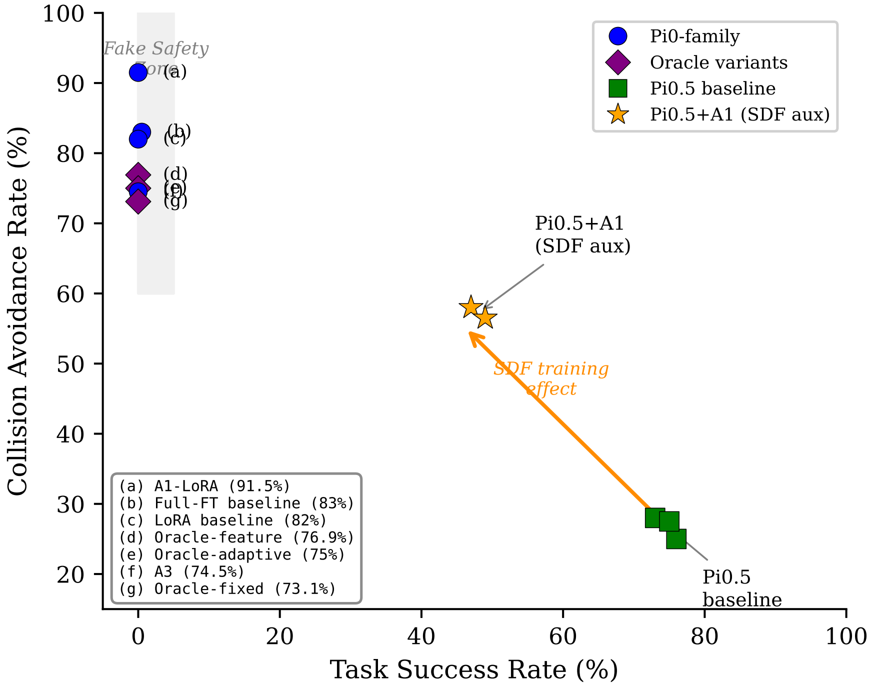

Figure 2: TSR vs. CAR for all configurations. Pi0-family models cluster in the upper-left (high CAR from inaction), Pi0.5 in the lower-right (high TSR with collision exposure), fine-tuned Pi0.5 in the center. The upper-right quadrant (high TSR and CAR) represents genuinely safe behavior. Error bars: ± 1 std across seeds.

Table 3: Oracle SDF refinement with ground-truth distance fields. All oracle variants achieve lower CAR than unrefined baselines. Per-task breakdown in Appendix F.

Method	CAR (%)	Fake Safe (%)	Refines/ep
LoRA baseline (no refine)	82.0	99.4	–
Full-FT baseline (no refine)	83.0	–	–
Oracle-adaptive (GT, dyn. α)	75.0	98.3	14.2
Oracle-fixed (GT, α=0.005)	73.1	98.3	14.8
Oracle-feature (GT, feat. space)	76.9	96.7	241.7
A3 learned SDF	77.1	–	0.0

5 End-Effector-Only SDF Cannot Ensure Full-Body Safety

Could a perfect SDF signal enable effective EE-only refinement? We replace the learned SDF with ground-truth distance fields from the simulation geometry (§3.3). Table 3 and Figure 3 show that all three oracle variants achieve *lower* CAR than unrefined baselines: the best oracle (76.9%) falls 5.1 points below LoRA (82.0%). Refinement with perfect distance information produces more collisions, not fewer.

Per-task analysis reveals the mechanism (Appendix F). On the one task where the EE is the primary collision body, oracle refinement helps (+9 points). On the remaining three, the SDF gradient pushes the EE away from obstacles but rotates intermediate arm links into obstacle volumes; the net effect is one collision traded for another. The A3 learned SDF never triggers refinement (0 steps per episode). EE-only refinement is structurally counterproductive: joint-space coupling between EE displacement and arm-link rotation redistributes collisions to unmonitored links regardless of signal quality.

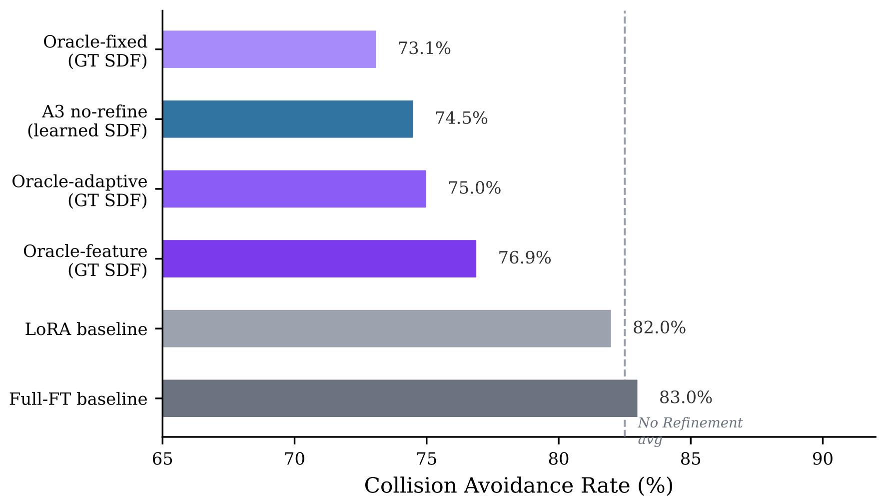

Figure 3: CAR for oracle SDF refinement versus unrefined baselines. All GT oracle variants underperform baselines despite perfect distance information.

Table 4: Avoidance source attribution on SafeLIBERO. SDF adds no measurable CAR beyond LoRA-only across the full tested range $\lambda \in [0.005, 0.1]$ (all $p > 0.05$). All fine-tuned models exhibit 0% frozen rate and displacement > 0.37 m.

Method	Seeds	TSR (%)	CAR (%)	EE Disp (m)
Pi0.5 baseline	3	74.7 ± 1.5	26.8 ± 1.6	0.367
LoRA-only (no SDF)	3	51.5 ± 0.0	54.67 ± 2.02	0.391
LoRA+SDF λ=0.005	3	54.0 ± 3.0	54.5 ± 2.29	0.387
LoRA+SDF λ=0.01	3	53.8 ± 1.5	56.33 ± 1.61	0.389
LoRA+SDF λ=0.1	3	50.2 ± 3.9	57.00 ± 2.18	0.394

6 Avoidance Source Diagnosis

Fine-tuning on collision-free demonstrations produces genuine collision avoidance: LoRA-adapted Pi0.5 raises CAR by ∼28 percentage points with zero frozen episodes. Does SDF geometric supervision contribute additional avoidance beyond what demonstrations provide? We compare LoRA-only against LoRA+SDF across a 20× range of loss weights ($\lambda \in \{0.005, 0.01, 0.1\}$) on Pi0.5, all evaluated at the 30K-step checkpoint.

Table 4 shows that collision-free demonstrations alone raise CAR by ∼28 percentage points (26.8% → 54.67%) with zero frozen episodes and displacement above 0.37 m, confirming genuine avoidance. Adding SDF supervision produces no measurable benefit across the full tested range: LoRA-only CAR (54.67%) is statistically indistinguishable from all SDF conditions including $\lambda = 0.1$ (57.00%, $p = 0.131$; Appendix G). A Phase 2 ablation ($\lambda = 0$ after step 5K) yields CAR = 57.5%[3], confirming that SDF gradients produce no lasting behavioral effect. The per-task CAR ordering is preserved across all conditions (chocolate_pudding highest, bbq_sauce lowest) and invariant to λ across a 20× range, indicating that avoidance patterns reflect the demonstration distribution rather than geometric supervision.

CAR–TSR trade-off. Fine-tuning degrades TSR by ∼21–23 pp (74.7% → 51.5–54.0%), a consequence of distributional shift from the pre-trained prior rather than learned paralysis (all models retain > 0.37 m displacement and 0% frozen episodes). The SSR/TSR ratio increases from 0.39

[3]Single seed (s2); computational constraints precluded additional seeds for this ablation.

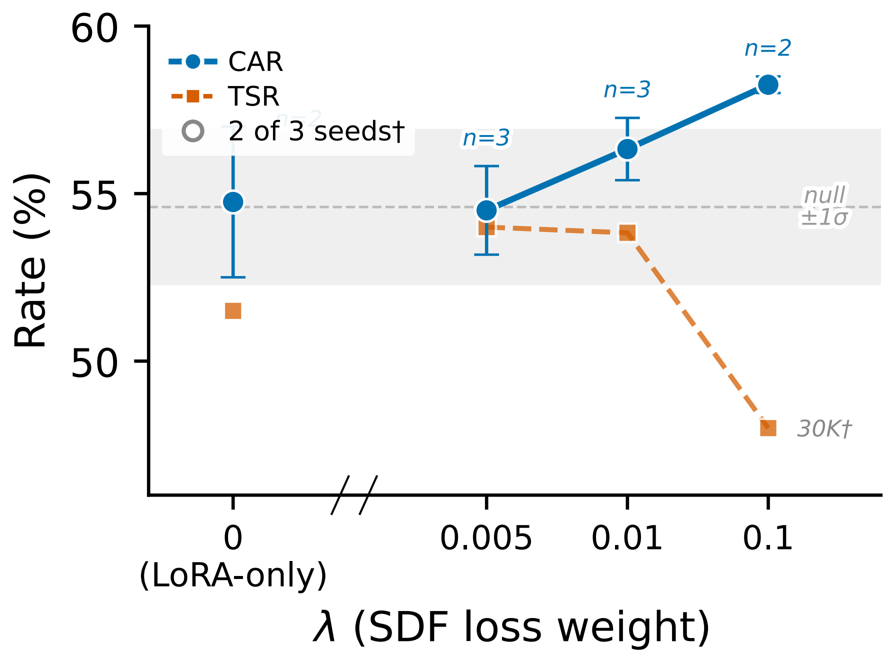

Figure 4: Effect of SDF loss weight λ on CAR and TSR. $\lambda = 0$: LoRA-only. All tested λ values remain within the null band ($p > 0.05$), confirming full redundancy of EE-only SDF supervision across a 20× loss-weight range.

(baseline) to 0.51–0.57 (Appendix E): among task-completing episodes, a larger fraction avoids collisions after fine-tuning.

Dose-response at higher loss weights. Increasing λ by 20× to 0.1 yields CAR = 57.00% ± 2.18 at the 30K checkpoint (3 seeds: 58.0%, 58.5%, 54.5%), within the null distribution (mean 54.58%, $\sigma = 1.93$; permutation $p = 0.131$[4]). SDF supervision is redundant with the demonstration signal across the full tested range $\lambda \in [0.005, 0.1]$ (Figure 4). This result disambiguates the two-level failure analysis (Section 7): because even a 20× increase in SDF loss weight fails to produce additional avoidance, the binding constraint is not training signal magnitude but the structural mismatch between end-effector monitoring and full-body collision surfaces. Stronger geometric supervision cannot compensate for supervising the wrong collision surface.

7 Discussion and Conclusion

Across 100+ experiments and 6000+ evaluation episodes on two VLA model families, three findings converge: current VLA safety evaluation is structurally unreliable. Action collapse renders reported CAR illusory (§3.2); EE-only oracle refinement is counterproductive (§5); and collision-free demonstrations, not geometric supervision, account for the measured avoidance gains (§6).

Why EE-only SDF fails. Two independent causes explain the null result. *(i) Demonstration data subsumes the SDF signal.* Collision-free demonstrations implicitly encode spatial avoidance; the ∼28 pp CAR gain arises primarily from this distributional shift, with per-task ordering invariant across all SDF conditions (§6). Even at $\lambda = 0.1$ (20× the lowest tested weight), CAR remains within the null distribution ($p = 0.131$), confirming that Level A signal masking is not the binding constraint—the failure is architectural. *(ii) EE monitoring cannot prevent full-body collisions.* The

[4]One-tailed permutation $p = 0.071$; even with the directional prior that SDF should not *decrease* CAR, the result remains non-significant.

SDF gradient acts on a single point while collisions occur along intermediate links. Oracle experiments confirm this: ground-truth SDF refinement *reduces* CAR by 6–10 pp below unrefined baselines (§5), because pushing the EE away rotates unmonitored links through obstacles. We hypothesize that collision-free demonstration trajectories implicitly encode workspace priors through their spatial distribution: the robot learns to favor regions of the workspace that are inherently collision-free, without requiring explicit geometric supervision. These two causes are independent; effective geometric safety requires both full-body supervision and training signals beyond demonstrations.

Implications for evaluation. dfCAR corrects action-collapse contamination using only EE position logging, already available in most evaluation pipelines. The bimodal displacement distribution makes τ_d robust across five orders of magnitude. We recommend reporting both CAR and dfCAR; the gap quantifies contamination. This contamination spans a 4000$\times$ parameter range (Pi0 Full-FT $\sim$2.5B to Pi0+A1 $\sim$658K trainable), a structural consequence of evaluating avoidance without verifying motion. More broadly, safety may need to operate at the planning or joint-space constraint layer, since any EE-centric loss inherits the same blind spot.

Limitations. Our evaluation covers one benchmark suite (SafeLIBERO, four tasks) and two model families (Pi0, Pi0.5). The tabletop setting with fixed-base manipulator and known obstacles may not generalize to mobile platforms and cluttered scenes. Oracle experiments used a largely frozen model (Pi0+A1, TSR$\approx$0%); testing on an actively moving policy would better isolate the structural limitation. Full-body SDF remains untested. Fine-tuning degrades TSR by $\sim$21–23 pp, yet SSR/TSR increases from 0.39 to 0.51–0.57 (Appendix E); reduced exposure alone does not explain the pattern.

Future work. Three directions follow: (1) full-body SDF supervision along the kinematic chain; (2) cross-architecture evaluation on OpenVLA, RT-2, and other VLA families; (3) planning-layer safety constraints in joint space rather than EE space.

Broader impact. Contaminated CAR scores allow models with no genuine avoidance to pass safety benchmarks. dfCAR exposes this gap with minimal changes to existing protocols.

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A Displacement Distribution

B Displacement Threshold Sensitivity Analysis

To validate the robustness of our displacement threshold τ_d, we evaluate the frozen rate across four orders of magnitude. Table 5 shows that the fake safety pattern persists across all reasonable threshold choices: A1-LoRA exhibits near-total action collapse regardless of threshold, while Pi0.5+A1 maintains 0% frozen rate even at the most conservative setting.

Table 5: Frozen rate (% of episodes with $d_{\max} < \tau_d$) across threshold values.

τ_d	A1-LoRA (Pi0)	Pi0.5 Baseline	Pi0.5+A1 (SDF aux)
0.001 mm (1 μm)	73.9	0.0	0.0
0.1 mm	99.6	0.0	0.0
1.0 mm	100.0	0.0	0.0
10 mm	100.0	0.0	0.0

The choice of $\tau_d = 0.1$ mm represents a conservative threshold: displacement below this value is physically negligible in manipulation tasks where typical action magnitudes are 1–10 mm per step. Even at $\tau_d = 1\,\mu$m, 73.9% of A1-LoRA safe episodes are classified as frozen, while all Pi0.5 episodes remain above threshold at every level tested.

Table 6 extends this analysis to dfCAR across fine-tuned Pi0.5 configurations. All eight model-seed combinations yield identical dfCAR for $\tau_d \in [2, 200]$ mm. At $\tau_d = 300$ mm, 0–2 episodes per seed

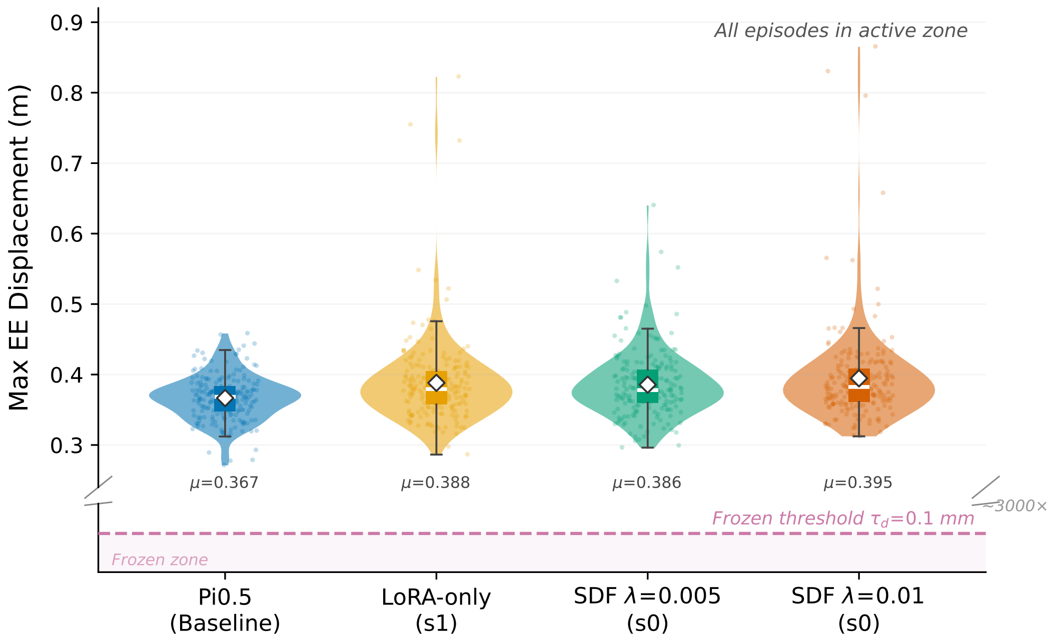

Figure 5: Displacement distribution across evaluation episodes. Pi0-family models (blue) cluster below 1 μm, while Pi0.5 episodes (orange) exceed 200 mm. The five-order-of-magnitude gap makes τ_d robust to any value in $[2, 200]$ mm.

fall below threshold (displacement 200–300 mm), causing $\leq$0.5 pp deviation. Our conclusions hold across the entire tested range.

Table 6: dfCAR (%) across displacement thresholds for fine-tuned Pi0.5 configurations ($n = 200$ episodes per seed). Values are identical across $\tau_d \in [2, 200]$ mm; at $\tau_d = 300$ mm, at most 2 episodes are excluded.

	τ_d (mm)							
Condition	2	5	10	20	50	100	200	300
LoRA s0	56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5
LoRA s1	52.5	52.5	52.5	52.5	52.5	52.5	52.5	52.5
SDF λ=0.005 s0	57.0	57.0	57.0	57.0	57.0	57.0	57.0	57.1
SDF λ=0.005 s1	54.0	54.0	54.0	54.0	54.0	54.0	54.0	54.0
SDF λ=0.005 s2	52.5	52.5	52.5	52.5	52.5	52.5	52.5	52.3
SDF λ=0.01 s0	57.0	57.0	57.0	57.0	57.0	57.0	57.0	57.0
SDF λ=0.01 s1	57.5	57.5	57.5	57.5	57.5	57.5	57.5	57.1
SDF λ=0.01 s2	54.5	54.5	54.5	54.5	54.5	54.5	54.5	54.0

C Per-Task TSR Breakdown

Table 7 decomposes TSR by task for the avoidance source attribution experiments (Section 6). All tasks retain meaningful success rates after fine-tuning, with Chocolate Pudding most resilient and Orange Juice most affected.

D Per-Task CAR Breakdown

Table 8 decomposes CAR by task for all fine-tuned conditions, supporting the claim that per-task ordering is preserved across SDF loss weights (§6). Chocolate Pudding is consistently highest and BBQ Sauce consistently lowest across all conditions, including $\lambda = 0.1$. The intermediate tasks

Table 7: Per-task TSR (%) at 30K training steps. Baseline reports the unified seed-7 re-evaluation (aggregate TSR = 71.5%). LoRA-only reports 3-seed means (re-evaluated with unified seed-7 protocol); SDF conditions report 3-seed means.

Task	Pi0.5 Base (seed = 7)	LoRA-only (3 seeds)	SDF λ=0.005 (3 seeds)	SDF λ=0.01 (3 seeds)
Orange Juice	74.0	30.0	33.3	24.7
Choco. Pudding	80.0	70.7	78.0	76.7
Milk	66.0	44.7	35.3	48.7
BBQ Sauce	66.0	60.7	69.3	65.3
Mean	71.5	51.5	54.0	53.8

(Orange Juice and Milk) exchange rank between seeds but remain bounded by the extremes. This invariance indicates that avoidance patterns reflect the spatial statistics of collision-free demonstrations rather than geometric supervision.

Table 8: Per-task CAR (%) at 30K steps. λ=0.005 and λ=0.01 report 3-seed means. Per-task ordering (CP highest, BBQ lowest) is preserved across all conditions.

Condition	OJ	CP	Milk	BBQ	Agg.
LoRA s0	56	74	70	26	56.5
LoRA s1	54	68	72	16	52.5
LoRA s2	56	76	62	26	55.0
SDF λ=0.005 (3-seed)	54.0	72.7	71.3	20.0	54.5
SDF λ=0.01 (3-seed)	57.3	72.7	72.0	23.3	56.3
SDF λ=0.1 s0	60	74	68	30	58.0
SDF λ=0.1 s1	66	72	70	26	58.5

For the $\lambda = 0.1$ condition, seed 1 records `avg_max_obstacle_displacement` of 0.0553 m (per-task: OJ = 0.033, CP = 0.055, Milk = 0.041, BBQ = 0.093), comparable to other fine-tuned conditions and consistent with active manipulation rather than frozen behavior.

E Safe Success Rate Analysis

Table 9 reports the Safe Success Rate (SSR = TSR $\cap$ CAR), the fraction of episodes achieving both task completion and collision avoidance. The SSR/TSR ratio quantifies the proportion of successful episodes that are also collision-free, controlling for task-completion rate.

Table 9: SSR and SSR/TSR ratio across conditions. The ratio increase from 0.39 (baseline) to 0.51–0.56 (fine-tuned) indicates that CAR gains reflect learned safer execution, not reduced task exposure.

Condition	TSR (%)	CAR (%)	SSR (%)	SSR/TSR
Pi0.5 baseline	71.5	28.5	28.0	0.39
LoRA-only (3 seeds)	51.5	54.67	27.7	0.54
SDF λ=0.005 (3 seeds)	54.0	54.5	27.8	0.51
SDF λ=0.01 (3 seeds)	53.8	56.33	30.2	0.56
SDF λ=0.1†	48.0	58.25	26.5	0.55

†SSR from seed 1 only; seed 0 per-episode data overwritten.

F Per-Task Oracle Analysis

Table 10 shows per-task CAR for oracle refinement. On task 1 (`chocolate_pudding`), where the EE is the primary collision body and the LoRA baseline achieves only 46.0% CAR, Oracle-adaptive improves to 55.0% and Oracle-feature to 47.5%. On the remaining tasks, all oracle variants degrade:

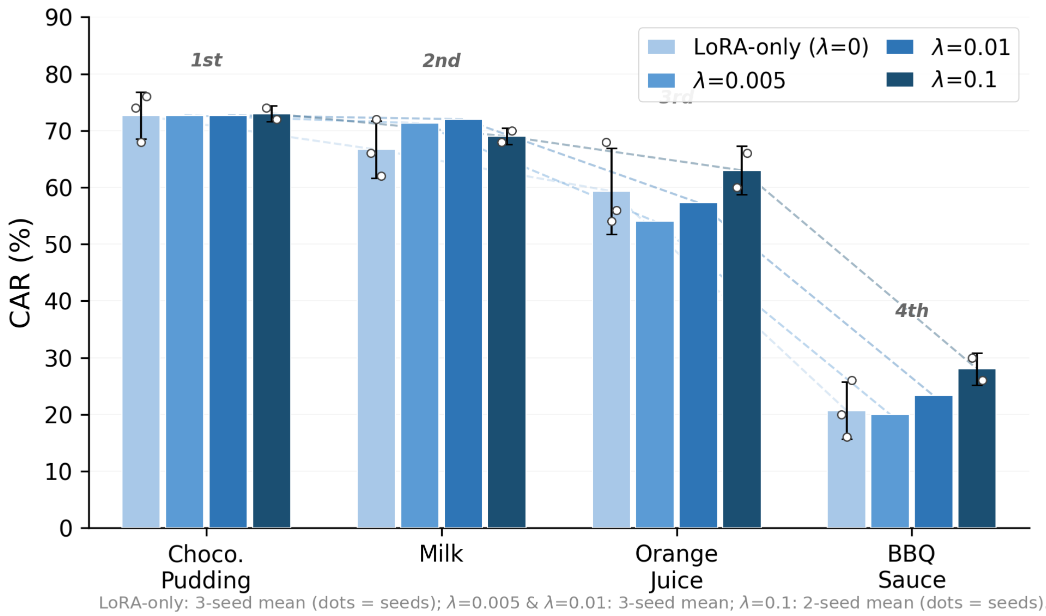

Figure 6: Per-task CAR fingerprint across conditions. Each bar shows the 3-seed mean; dots indicate individual seeds. The task ordering (chocolate_pudding highest, bbq_sauce lowest) is preserved across all λ values, confirming that avoidance patterns reflect the demonstration distribution rather than geometric supervision. LoRA-only: 3-seed mean.

Oracle-adaptive reduces task 0 (`orange_juice`) from 96.0% to 82.5% and task 2 (`milk`) from 98.0% to 77.5%. The SDF gradient pushes the EE away from the nearest surface, but the resulting joint-space displacement rotates intermediate arm links into obstacle volumes.

The feature-space oracle fires on nearly every timestep (241.7 refinements/episode vs. 14.2 for Cartesian variants) but with gradient magnitudes two orders of magnitude smaller (0.004–0.006 vs. 0.18–0.49), and CAR (76.9%) shows no advantage despite $17\times$ more steps. The A3 learned SDF never triggers refinement (0 steps/episode), as its outputs avoid the activation threshold.

Table 10: Per-task CAR (%) for oracle SDF refinement variants. EE-only refinement helps only on task 1 where the EE is the primary collision body; on other tasks it trades one collision site for another.

Method	Task 0	Task 1	Task 2	Task 3	Agg.
LoRA baseline	96.0	46.0	98.0	88.0	82.0
Oracle-adaptive	82.5	55.0	77.5	85.0	75.0
Oracle-fixed	85.0	50.0	77.5	80.0	73.1
Oracle-feature	90.0	47.5	82.5	87.5	76.9

G Null Distribution for SDF Loss Weight Analysis

We construct a null distribution by pooling the three LoRA-only seeds and three $\lambda = 0.005$ seeds into six per-seed CAR measurements: 56.5% (LoRA s0, re-evaluated), 52.5% (LoRA s1), 55.0% (LoRA s2), 57.0% (λ=0.005 s0), 54.0% (λ=0.005 s1), 52.5% (λ=0.005 s2), yielding mean 54.58% and standard deviation 1.93% (Bessel-corrected, $n-1 = 5$). We treat these as exchangeable because SDF at $\lambda = 0.005$ produces no separation from LoRA-only. The $\lambda = 0.01$ mean of 56.33% falls within 0.8σ of this null ($p > 0.1$), consistent with seed-to-seed variation rather than a treatment effect. The $\lambda = 0.1$ 30K mean of 58.25% (2 seeds) yields a two-tailed permutation $p \approx 0.071$ (one-tailed $p = 0.036$); a z-test gives an approximate $p \approx 0.039$ ($z = 1.76$). This marginal departure is inconclusive at $n = 2$.

The permutation test makes no distributional assumptions. Pooling the six null values $\{56.5, 52.5, 55, 57, 54, 52.5\}$ and two $\lambda = 0.1$ values $\{58, 58.5\}$ into eight measurements, we enumerate all $\binom{8}{2} = 28$ two-element combinations and compute the fraction whose mean ≥ 58.25. Only one combination ($\{58, 58.5\}$, mean $= 58.25$) meets this threshold, yielding an exact one-tailed permutation $p = 1/28 \approx 0.036$ (two-tailed $p = 2/28 \approx 0.071$, as one combination on the lower tail also reaches equal distance from the pooled mean). The consistency of the exact permutation and approximate z-test results confirms that the marginal signal is robust to distributional assumptions but remains inconclusive at $n = 2$.

A Phase 2 ablation reinforces this conclusion. Setting $\lambda = 0$ during Phase 2 (steps 5K–30K), so that SDF supervision is active only during auxiliary head warmup (Phase 1, steps 0–5K), yields CAR of 57.5% (single seed, s2). This value lies within the null range, indicating that SDF training produces no lasting behavioral effect on collision avoidance.

H Step-wise Checkpoint Analysis for $\lambda = 0.1$

Table 11 reports per-seed CAR at seven checkpoints for $\lambda = 0.1$. Seed-1 data is available at 16K, 20K, 22K, and 30K; 24K–28K report seed-0 only. The seed-0 curve shows no monotonic trend through 28K; the 30K checkpoint marks the first consistent dual-seed departure from the null band, though whether this reflects a genuine late-training effect or sampling variance at $n = 2$ remains open.

Table 11: Per-seed CAR (%) at training checkpoints for $\lambda = 0.1$ (200 episodes each). 24K–28K: seed-0 only. Dual-seed means remain within the null band ($54.58\% \pm 1.93\%$) at all checkpoints except 30K, where both seeds depart marginally (two-tailed permutation $p \approx 0.071$).

Checkpoint	Seed 0	Seed 1	Mean
16K steps	55.5	56.0	55.75
20K steps	55.0	56.0	55.5
22K steps	60.5	56.5	58.5
24K steps	52.5	–	–
26K steps	53.5	–	–
28K steps	54.5	–	–
30K steps	58.0	58.5	58.25

Individual seeds exhibit checkpoint-to-checkpoint CAR fluctuations of up to 8 pp (seed 0: 52.5–60.5%), larger than the cross-seed standard deviation at any fixed checkpoint ($\leq$2.8 pp). This is consistent with binomial sampling noise at $n = 50$ episodes per task ($\sigma \approx 7$ pp per task for $p \approx 0.5$). The seed-averaged curve shows no monotonic trend, confirming that SDF loss weight does not produce a progressive training signal.

I Oracle SDF Refinement Details

All oracle variants replace the learned SDF decoder with ground-truth distance fields computed from the simulation geometry. Each is evaluated for 160 episodes (4 tasks $\times$ 40).

Oracle-adaptive. At each timestep, if the ground-truth SDF at the end-effector position falls below a safety margin, the action is adjusted by a step in the negative SDF gradient direction. The step size α is scaled inversely with distance: smaller distances produce larger corrections. This mimics an adaptive potential field controller.

Oracle-fixed. Identical to Oracle-adaptive except the step size is fixed at $\alpha = 0.005$ regardless of distance. This isolates the effect of step-size adaptation from the refinement geometry.

Oracle-feature. Instead of modifying the action directly in Cartesian space, this variant applies the SDF gradient in the policy's latent feature space via backpropagation. The gradient of the ground-truth SDF value with respect to the intermediate features is computed and used to perturb the features before the action head decodes them. This tests whether geometric corrections are more effective when applied in the model's learned representation space. In practice, the feature-space variant fires on nearly every timestep (241.7 refinements per episode vs. 14.2 for Cartesian variants) but with gradient magnitudes two orders of magnitude smaller (0.004–0.006 vs. 0.18–0.49 in action space).

J Training Details

Architecture. We use LoRA [31] adapters injected into the PaliGemma vision encoder (rank 16, α=16) and the action expert (rank 32, α=32), following the configuration of OpenPi. The SDF decoder is a 4-layer MLP (hidden dimension 256) with skip connections, taking pooled suffix features (1024-d) and query points (3-d) as input. The end-effector extractor is a 2-layer MLP (1024 $\rightarrow$ 256 $\rightarrow$ 3).

Training schedule. Training proceeds in two phases:

- **Phase 1** (steps 0–5000): Backbone and LoRA parameters are frozen. Only the SDF decoder and EE extractor are trained, allowing them to learn meaningful representations before the policy adapts.
- **Phase 2** (steps 5000–30000): LoRA parameters are unfrozen with a 100-step gradient warmup ramp. All auxiliary heads continue training jointly with the policy.

Table 12: Training hyperparameters.

Parameter	Value
Base model	Pi0.5 (OpenPi, $\sim$3.5B parameters)
Optimizer	AdamW
Learning rate	2.5×10^{-5} (cosine decay to $\mathrm{lr}/10$)
Warmup steps	1000
Batch size	32
Total steps	30000
λ_{sdf}	$\{0.005, 0.01, 0.1\}$
λ_{ee}	0.01
SDF query points per step	64
SDF clamp δ	0.05 m
Eikonal weight	0.1
LoRA rank (vision)	16
LoRA rank (action)	32
Phase 1 duration	5000 steps

Hyperparameters.

Evaluation protocol. Each episode runs for at most 300 timesteps. The policy generates a 50-step action trajectory via flow matching and executes the first 5 steps before replanning. An episode terminates early upon task completion. Collision detection uses MuJoCo's built-in contact detection: any contact between the robot and a non-target object registers as a collision. For Pi0.5 evaluations, camera images are rotated 180° to match the pre-training viewpoint convention; this correction was verified to restore TSR from $\sim$5% to $\sim$52% while leaving CAR comparable. Oracle variants use 40 episodes per task (160 total) with a single seed; all other configurations use 50 episodes per task (200 total).

Data. We use the SafeLIBERO object_pudding suite with pre-computed SDF ground truth. SDF values are computed offline from the obstacle mesh using a dense 64^3 voxel grid. During training, 64 query points are randomly sampled per step from this grid.

Computational resources. All training and evaluation experiments are conducted on NVIDIA RTX PRO 6000 GPUs (96 GB). Phase 1 (5K steps) takes approximately 6 hours on a single GPU; Phase 2 (25K steps) takes approximately 32 hours. Evaluation of 200 episodes requires approximately 4 hours. Across 100+ experiments (training runs, evaluations, oracle variants, and ablations), the total computational budget is approximately 700 GPU-hours.